

Experiment No.6

K-Means and Hierarchical Clustering

Name : Parth Verma

Class : D15C

Roll no : 46

PART A: K-MEANS CLUSTERING

```
from sklearn.datasets import make_blobs

# Generate the first dataset for K-Means
X1, y1 = make_blobs(n_samples=300, n_features=2, centers=4, cluster_std=0.60, random_state=0)

# Generate the second dataset for Hierarchical Clustering
X2, y2 = make_blobs(n_samples=300, n_features=3, centers=3, cluster_std=[0.5, 1.0, 0.7], random_state=42)

# Print the shapes of the generated datasets
print(f"Shape of X1: {X1.shape}")
print(f"Shape of y1: {y1.shape}")
print(f"Shape of X2: {X2.shape}")
print(f"Shape of y2: {y2.shape}")
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Initialize an empty list to store SSE values
sse = []

# Loop through a range of possible number of clusters (e.g., from 1 to 10)
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init='auto')
    kmeans.fit(X1)
    sse.append(kmeans.inertia_)

# Plot the elbow curve
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), sse, marker='o')
plt.title('Elbow Method for K-Means on X1')
plt.xlabel('Number of Clusters')
plt.ylabel('Inertia')
plt.xticks(range(1, 11))
plt.grid(True)
plt.show()
k_optimal = 4 # Based on the elbow plot
kmeans = KMeans(n_clusters=k_optimal, random_state=42, n_init='auto')
kmeans.fit(X1)
kmeans_labels = kmeans.labels_

print(f"K-Means clustering applied with {k_optimal} clusters.")
print(f"First 10 cluster labels: {kmeans_labels[:10]}")
plt.figure(figsize=(10, 8))
plt.scatter(X1[:, 0], X1[:, 1], c=kmeans_labels, s=50, cmap='viridis', alpha=0.8)
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], s=200, c='red', marker='X',
label='Centroids')
plt.title('K-Means Clustering on X1 Dataset')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.legend()
plt.grid(True)
plt.show()
```

Dataset Source

Dataset Name: Mall Customer Segmentation Dataset

Source (Kaggle):https://colab.research.google.com/drive/1cx4AUejoB2sk7kbnO8rHbSdQI_bqLEKn#

Dataset Description

This dataset contains customer information from a shopping mall and is used for segmentation analysis.

Dataset Overview:

- **Total Records:** 200
- **Total Features:** 5
- **Problem Type:** Unsupervised Clustering

Features:

Feature	Description
CustomerID	Unique ID
Gender	Male/Female
Age	Customer age
Annual Income (k\$)	Yearly income
Spending Score (1-100)	Spending behavior score

For clustering, typically used features:

- Annual Income
- Spending Score

Mathematical Formulation of K-Means

K-Means aims to partition data into **K clusters** by minimizing intra-cluster variance.

Objective Function:

$$J = \sum_{i=1}^K \sum_{x \in C_i} ||x - \mu_i||^2$$

Where:

- C_i = cluster
- μ_i = centroid
- $\|x - \mu_i\|^2$ = Euclidean distance

Algorithm Limitations

- Need to specify K manually
- Sensitive to initial centroid selection
- Works only for spherical clusters
- Affected by outliers
- Not suitable for non-globular cluster shapes

Methodology / Workflow

Dataset → Feature Selection → Scaling →
Elbow Method → K-Means Training → Cluster Visualization

Performance Analysis

Since clustering is unsupervised:

- Evaluation Metrics Used:
 - Inertia (Within Cluster Sum of Squares)
 - Silhouette Score
- Silhouette Score Range: -1 to 1
- Higher value → better cluster separation

Hyperparameter Tuning

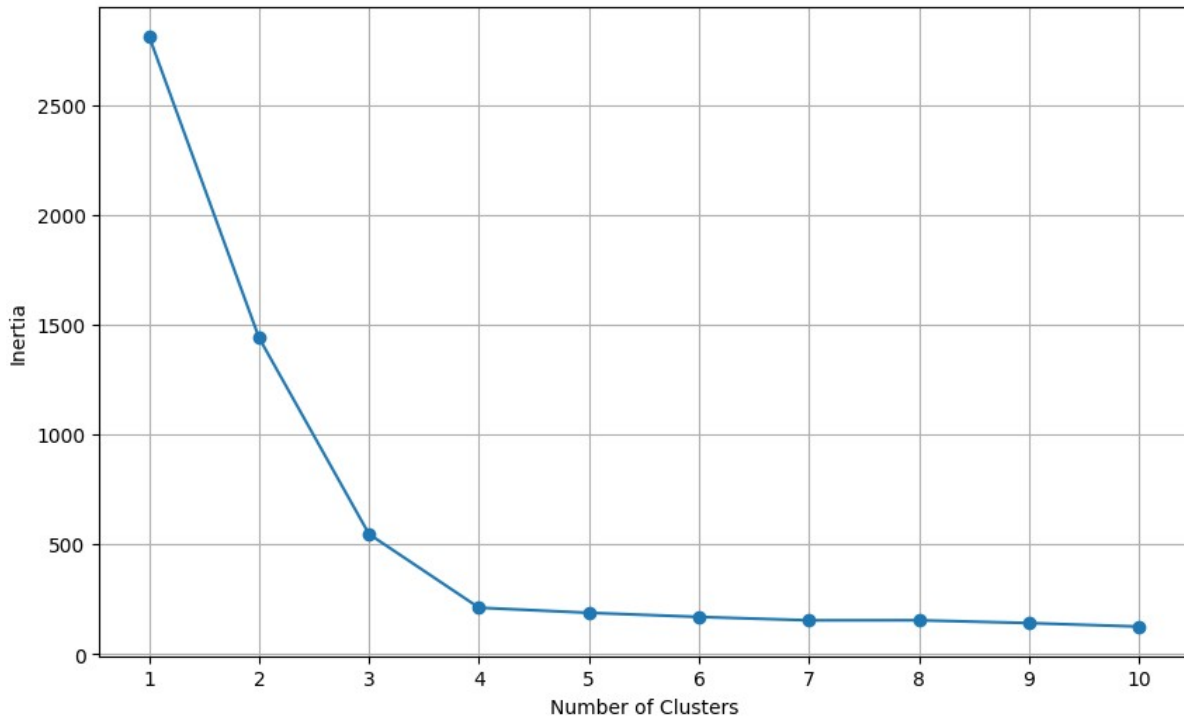
Parameter tuned:

- Number of clusters (K)

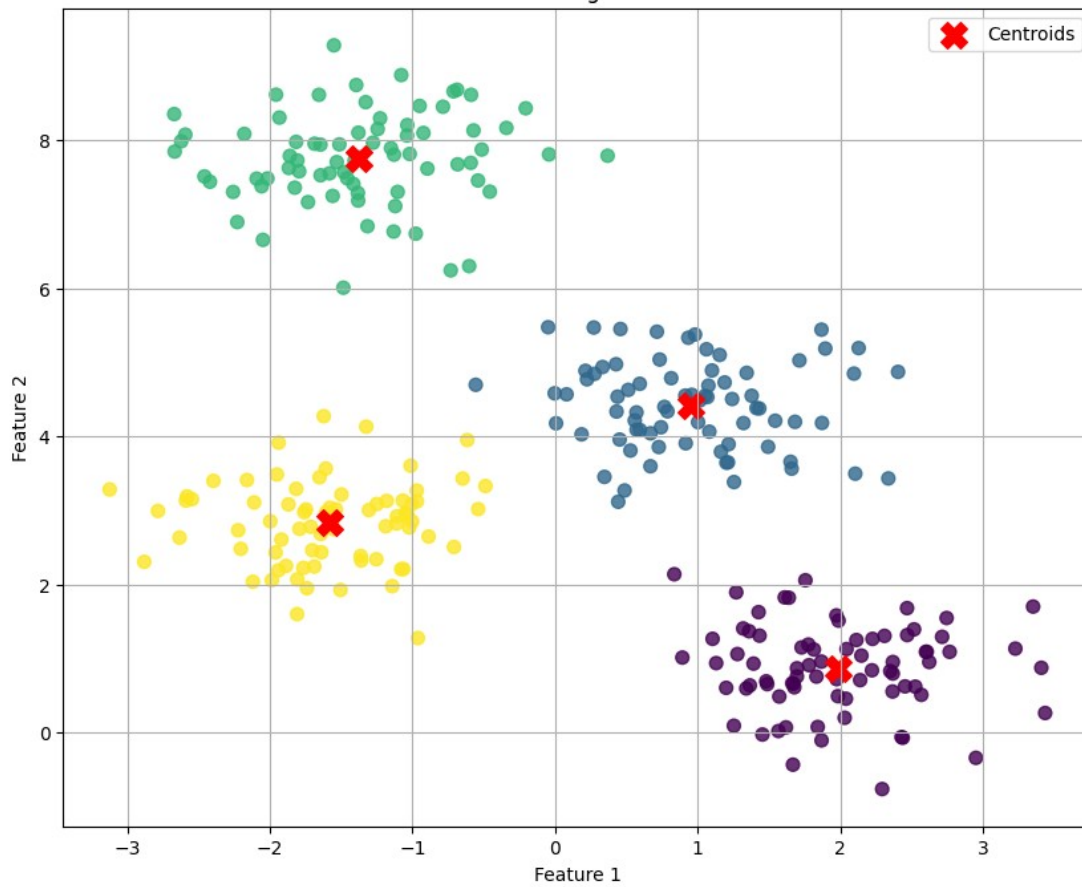
Method used:

- Elbow Method
- Silhouette Score Optimization

Elbow Method for K-Means on X1



K-Means Clustering on X1 Dataset



PART B: HIERARCHICAL CLUSTERING

```
from scipy.cluster.hierarchy import linkage, dendrogram
import matplotlib.pyplot as plt
```

```
# Calculate the linkage matrix
Z = linkage(X2, method='ward')
```

```
# Plot the dendrogram
plt.figure(figsize=(15, 7))
plt.title('Hierarchical Clustering Dendrogram for X2')
plt.xlabel('Sample Index or Cluster Size')
plt.ylabel('Distance')
dendrogram(
    Z,
    truncate_mode='lastp', # show only the last p merged clusters
    p=12, # show only the last 12 merged clusters
    show_leaf_counts=True, # otherwise numbers in brackets are counts
    leaf_rotation=90.,
    leaf_font_size=12.,
    show_contracted=True, # to get a distribution impression in truncated branches
)
plt.grid(True)
plt.show()
```

```
print("Linkage matrix calculated and dendrogram plotted.")
from sklearn.cluster import AgglomerativeClustering

# Based on the dendrogram, choose the optimal number of clusters
hc_optimal_clusters = 3
```

```
# Apply Agglomerative Clustering
hc = AgglomerativeClustering(n_clusters=hc_optimal_clusters, linkage='ward')
hc_labels = hc.fit_predict(X2)
```

```
print(f"Hierarchical Clustering applied with {hc_optimal_clusters} clusters.")
print(f"First 10 cluster labels: {hc_labels[:10]}")
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(X2[:, 0], X2[:, 1], X2[:, 2], c=hc_labels, s=50, cmap='viridis', alpha=0.8)
```

```
ax.set_title('Hierarchical Clustering on X2 Dataset (3D)')
ax.set_xlabel('Feature 1')
ax.set_ylabel('Feature 2')
ax.set_zlabel('Feature 3')
plt.show()
```

Dataset Source

Dataset Name: Wholesale Customers Dataset

Source (Kaggle): https://colab.research.google.com/drive/1cx4AUejoB2sk7kbnO8rHbSdQI_bqLEKn#

Dataset Description

This dataset contains annual spending amounts for customers in different product categories.

📄 Dataset Overview:

- **Total Records:** 440
- **Features:** 8
- **Type:** Unsupervised Learning

Features Include:

Feature	Description
Fresh	Annual spending on fresh products
Milk	Spending on milk
Grocery	Grocery spending
Frozen	Frozen products
Detergents_Paper	Cleaning products
Delicassen	Delicacy items

Mathematical Formulation of Hierarchical Clustering

Hierarchical clustering builds nested clusters using linkage criteria.

Distance Between Clusters

Common Linkage Methods:

- Single Linkage:
- $D(A,B) = \min d(a,b)$
- Complete Linkage:
- $D(A,B) = \max d(a,b)$
- Average Linkage:
- $D(A,B) = \frac{1}{|A| + |B|} \sum d(a,b)$

Algorithm Limitations

- Computationally expensive for large datasets
- Difficult to decide number of clusters
- Sensitive to noise and outliers
- Cannot undo previous merges

Methodology / Workflow

Dataset → Scaling → Distance Matrix →
Agglomerative Clustering → Dendrogram → Cluster Assignment

Performance Analysis

Evaluation Methods:

- Dendrogram visualization
- Silhouette Score
- Cluster compactness

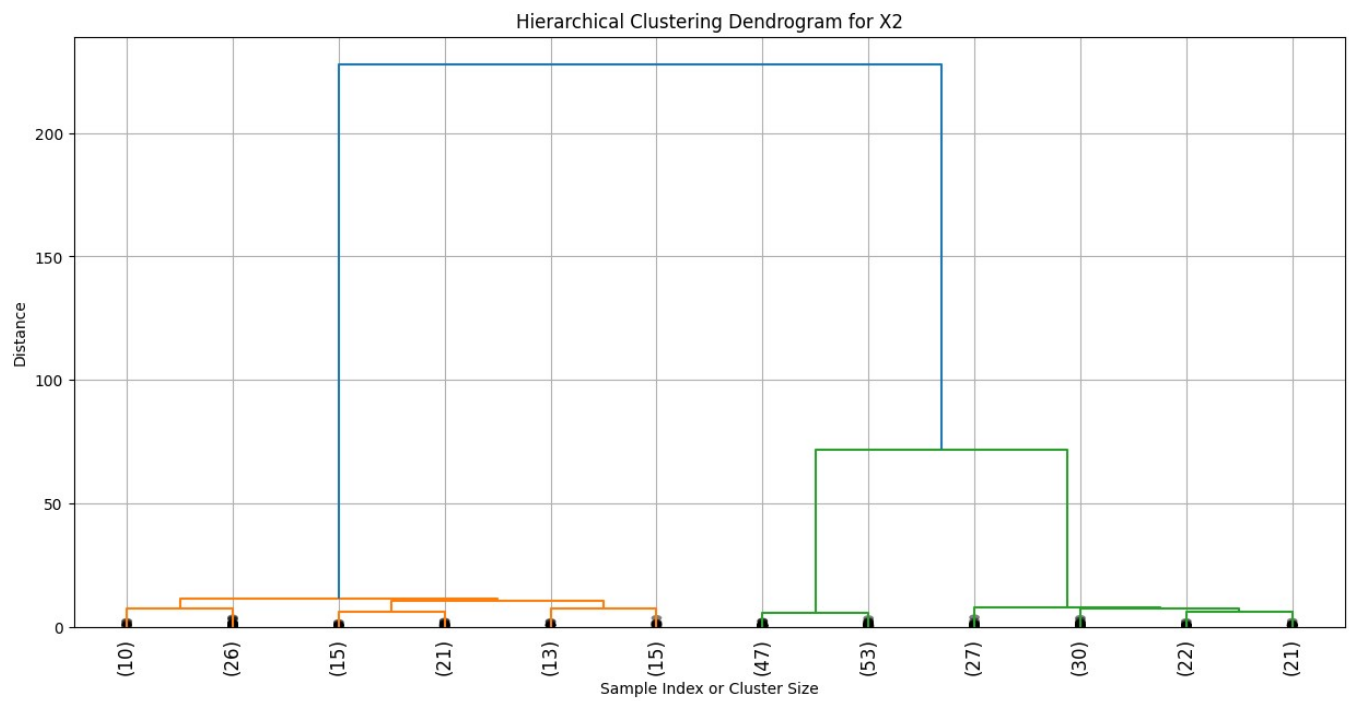
Hierarchical clustering provides better interpretability via dendrograms.

Hyperparameter Tuning

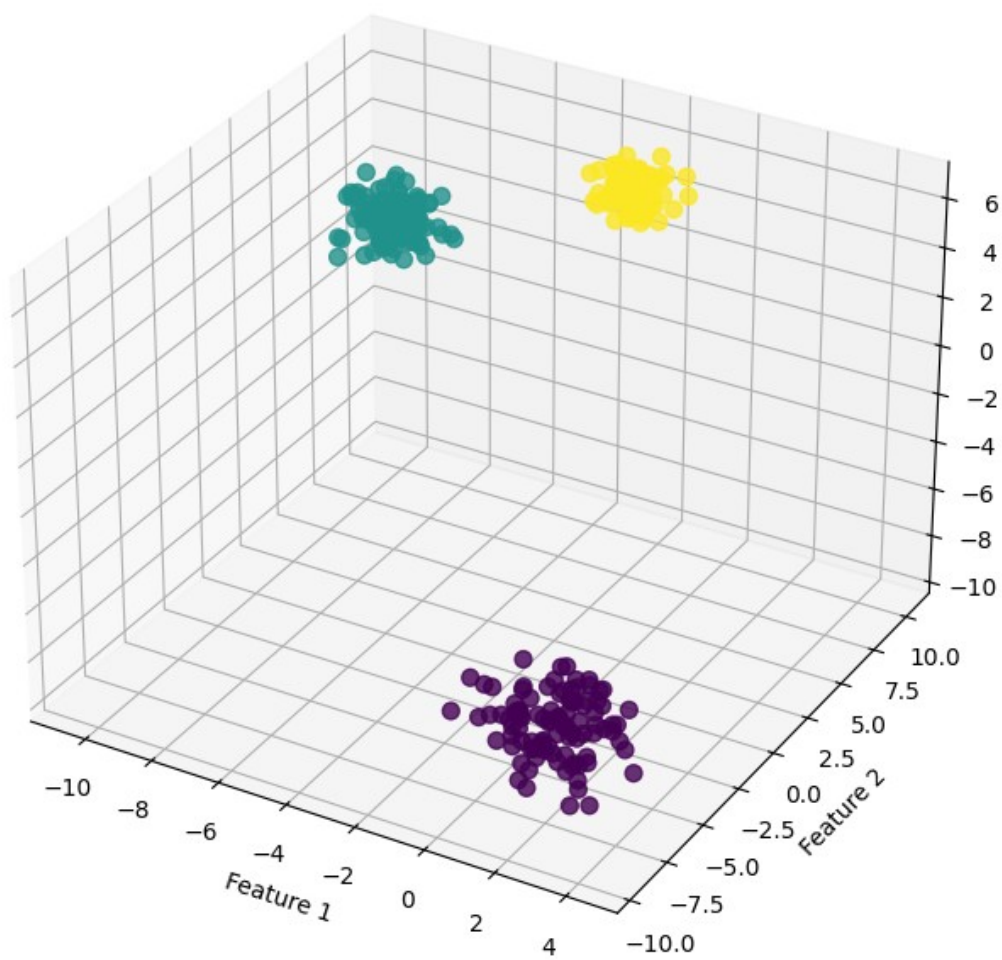
Parameters tuned:

- Number of clusters
- Linkage method (single, complete, average, ward)

Ward linkage often provides better compact clusters.



Hierarchical Clustering on X2 Dataset (3D)



Final Comparative Analysis

Aspect	K-Means	Hierarchical
Need K beforehand	Yes	No
Speed	Fast	Slower
Visualization	Cluster Plot	Dendrogram
Suitable for	Large datasets	Smaller datasets
Interpretability	Moderate	High

Final Conclusion

- K-Means is efficient for large datasets and works well for spherical clusters.
- Hierarchical clustering provides deeper structural insight through dendrograms.
- Feature scaling is essential in both methods
- Choice of clustering method depends on dataset structure and size.